ON A CONJECTURE OF ERDŐS OVER FUNCTION FIELDS

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Abstract. Based on Katz's equidistribution framework, we give a one-dimensional proof of a function-field analogue of a conjecture of Erdős in the large-q regime: for squarefree f of degree n, every class in $(\mathbb{F}_q[t]/(f))^\times$ is represented as a product of two monic irreducibles of degrees $\le n$ once q is sufficiently large in terms of n. The proof yields a uniform $q^{-1/2}$-saving in the relevant twisted character sums, with an explicit constant depending on n.

Sawin [13] proved a stronger and more general result by higher-dimensional sheaf-theoretic methods; the present note instead emphasizes a simpler one-dimensional argument within Katz's framework.

1. Introduction

Many central problems in additive and multiplicative number theory remain unresolved over the integers. A notable example is the twin prime conjecture, which asserts the infinitude of prime pairs differing by two. The strongest progress toward this conjecture, achieved through sieve methods, is due to Chen [2], who proved that there exist infinitely many primes p such that $p + 2$ is the product of at most two primes. By contrast, in the setting of global function fields, the availability of geometric techniques has led to major advances. In particular, Sawin and Shusterman [12] resolved the function field analogues of both Chowla's and the twin prime conjecture.

Erdős conjectured that, for every sufficiently large prime modulus q, each reduced residue class modulo q can be represented as a product p_1p_2 of two primes with $p_1, p_2 \le q$; see [4, §2]. Matomäki and Teräväinen [10] established a ternary version of this conjecture (representation by three primes) and proved the binary case for all but a small exceptional set of residue classes.

Let q be a prime power, and let $k = \mathbb{F}_q$ be a finite field with q elements. Let $f \in k[X]$ be a squarefree polynomial of degree $n \ge 2$. Set $B := k[X]/(f(X))$, and let $B^\times$ be the group of units of B. Define

$$E(k, f) := \Big\{ a \in B^\times : a \equiv f_1 f_2 \pmod{f},\ f_1, f_2 \in k[X] \text{ monic irreducible with } \deg(f_i) \le n \Big\}.$$

Lemma 9.14 of Sawin [13] yields a function field analogue of Erdős' conjecture: if $q > 1024e^2$ and n is sufficiently large, then every reduced residue class modulo any squarefree polynomial of degree n can be written as a product of two monic irreducible polynomials of degree n. Here the natural size parameter in the function field analogue of Erdős' conjecture is

$$|f| = q^{\deg f} = q^n,$$

which is taken to be sufficiently large.

Specializing Lemma 9.14 of Sawin [13] with the substitutions

$$c \mapsto 0, \quad n \mapsto 2n, \quad m \mapsto n, \quad g \mapsto f, \quad \omega \mapsto 2, \quad n_1 \mapsto n,\ n_2 \mapsto n, \quad h \mapsto 1,$$

the left-hand side of the lemma counts the number of pairs of monic irreducible polynomials (f_1, f_2) of degree n such that $f_1 f_2 \equiv a \pmod{f}$, minus a main term of size q^n. The error term appearing in Lemma 9.14 is

$$O\Big(\frac{(2n)!}{(n!)^2}(8e\sqrt{q})^n\Big) = O\left((32e\sqrt{q})^n\right),$$

which is smaller than the main term whenever $q > 1024e^2 \approx 7566.4$ and n is sufficiently large. Therefore $E(k, f) = B^\times$ holds for all sufficiently large n.

In this paper, we establish by a different method the function-field analogue of Erdős' conjecture in the large finite-field limit, using Katz's convolution–equidistribution framework [8, 9].

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Theorem 1.1. *Let $n \geq 2$. Then there exists a constant $Q(n) > 0$ such that, for every finite field $k = \mathbb{F}_q$ with $q \geq Q(n)$ and every squarefree polynomial $f \in k[X]$ of degree n, letting $B := k[X]/(f(X))$, we have*

$$E(k,f) := \left\{ a \in B^\times : a \equiv f_1 f_2 \pmod f,\ f_1, f_2 \in k[X] \text{ monic irreducible with } \deg(f_i) \leq n \right\} = B^\times.$$

Moreover, one may take $Q(n)$ to be of the form $n^{\kappa n}$ for some absolute constant $\kappa > 0$; for instance, $\kappa = 16$ suffices.

Corollary 1.2. *Let $n \geq 2$, and let $Q(n)$ be as in Theorem 1.1. Fix an integer $1 \leq r < 2n$. For every finite field $k = \mathbb{F}_q$ with $q \geq Q(n)$ and every squarefree polynomial $f \in k[X]$ of degree n, set $B := k[X]/(f(X))$. Then every $a \in B^\times$ can be represented modulo f in the form*

$$a \equiv f_1 f_2 \pmod f, \qquad f_1, f_2 \in k[X] \text{ monic irreducible}, \qquad \deg f_1 = r, \quad \deg f_2 = 2n - r.$$

Remark 1.3. The proof of Theorem 1.1 gives an error term of size $C(n)q^{n-1/2}$ with $C(n)$ explicit, which is sufficient in the large-q regime considered here. The difference between the $q^{-1/2}$-saving here and the $q^{-n/2}$-saving in [13] reflects the dimension of the geometric space governing the relevant Frobenius-trace estimates.

2. Initial set-up

2.1. **Weighted counting functions.** Let q be a prime power, and let $k = \mathbb{F}_q$ be a finite field with q elements. Fix an algebraic closure $\bar{k}$ of k. Let $f \in k[X]$ be a squarefree polynomial of degree $n \geq 2$, and set

$$B := k[X]/(f(X)),$$

which is a finite étale k-algebra of degree n. Denote by $B^\times$ the multiplicative group of B.

Multiplying f by a nonzero scalar in k does not change the ideal (f), and hence does not change $B = k[X]/(f)$. Indeed, if $a \in k^\times$, then for any $g, h \in k[X]$,

$$g \equiv h \pmod{af} \iff g \equiv h \pmod f.$$

Hence we may replace f by its monic scalar multiple and assume that f is monic. Write

$$f = \prod_{i=1}^r f_i$$

as a product of distinct monic irreducible polynomials, with $\deg f_i = n_i$. Then $B \cong \prod_{i=1}^r \mathbb{F}_{q^{n_i}}$, and hence

$$|B^\times| = \prod_{i=1}^r (q^{n_i} - 1) = q^n \prod_{i=1}^r (1 - q^{-n_i}).$$

In particular, $(q-1)^n \leq |B^\times| \leq q^n$.

Fix $d \geq 1$. For $a \in B^\times$, define

$$S(a,d) := \frac{1}{|B^\times|} \sum_{\chi \bmod f} \chi(a^{-1}) \Bigg(\sum_{\substack{\deg g \leq d \\ g \text{ monic}}} \chi(g)\,\Lambda(g) \Bigg)^2, \tag{2.1}$$

where the sum over χ runs over all multiplicative characters $\chi : B^\times \to \mathbb{C}^\times$, and Λ denotes the von Mangoldt function for $k[X]$ given by

$$\Lambda(h) = \begin{cases} \deg p, & \text{if } h = up^m \text{ for some monic irreducible } p \in k[X],\ u \in k^\times, m \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

By orthogonality of the characters of $B^\times$, we have

$$S(a,d) = \sum_{\substack{g_1, g_2 \in k[X] \text{ monic} \\ \deg g_1, \deg g_2 \leq d \\ g_1 g_2 \equiv a \pmod f}} \Lambda(g_1)\Lambda(g_2). \tag{2.2}$$

Thus $S(a,d)$ is a weighted count of representations of a modulo f as a product of two monic prime powers of degrees at most d.

Define

$$S_p(a,d) := \sum_{\substack{p_1,p_2\in k[X] \text{ monic irreducible} \\ \deg p_1, \deg p_2 \le d \\ p_1p_2\equiv a \pmod f}} \deg p_1 \, \deg p_2.$$

Then $S_p(a,d) > 0$ implies that a is represented modulo f as a product of two monic irreducible polynomials of degrees at most d.

Lemma 2.1. *For every $a \in B^\times$, we have*

$$0 \le S(a,d) - S_p(a,d) \le \begin{cases} 4d^2 q^{d/2}, & d < n, \\ 4d^2\Big(1 + \sum_{j=0}^{d-n} q^j\Big) q^{d/2}, & d \ge n. \end{cases}$$

In particular, when $d = n$,

$$0 \le S(a,n) - S_p(a,n) \le 8n^2 q^{n/2}.$$

Proof. The difference $S(a,d) - S_p(a,d)$ comes from those terms $\Lambda(g_1)\Lambda(g_2)$ in (2.2) for which at least one of g_1, g_2 is a proper prime power.

We first bound the total von Mangoldt weight of proper prime powers of degree at most d. By [11, Prop. 2.1],

$$\sum_{\substack{g \text{ monic} \\ \deg g = m}} \Lambda(g) = q^m.$$

Thus,

$$\sum_{\substack{p \text{ monic irreducible} \\ \deg p = m}} m \le q^m,$$

and we have

$$\sum_{\substack{p^k,\ k\ge 2 \\ p \text{ monic irreducible} \\ k \deg p \le d}} \deg p \le \sum_{k=2}^{d} \sum_{m \le d/k} q^m \le 2d\, q^{d/2}.$$

Now fix $b \in B$. We bound the number of monic polynomials $h \in k[X]$ with $\deg h \le d$ and $h \equiv b \pmod f$. There is at most one such polynomial of degree $< n$. Moreover, for each $j \ge 0$, there are at most q^j monic polynomials of degree $n + j$ in the residue class $b \pmod f$. Hence

$$\sum_{\substack{h\in k[X] \text{ monic} \\ \deg h \le d \\ h \equiv b \pmod f}} \Lambda(h) \le \begin{cases} d, & d < n, \\ d + \sum_{j=0}^{d-n} (n+j) q^j \le d\Big(1 + \sum_{j=0}^{d-n} q^j\Big), & d \ge n. \end{cases}$$

We now bound the contribution of those terms $\Lambda(g_1)\Lambda(g_2)$ in (2.2) for which at least one of g_1, g_2 is a proper prime power. Suppose first that g_1 is a proper prime power. Since $g_1 g_2 \equiv a \pmod f$ with $a \in B^\times$, the image of g_1 in B is invertible, and g_2 must lie in the fixed residue class $a g_1^{-1} \pmod f$. Therefore the total contribution from such pairs is at most

$$\begin{cases} 2d^2 q^{d/2}, & d < n, \\ 2d^2\Big(1 + \sum_{j=0}^{d-n} q^j\Big) q^{d/2}, & d \ge n. \end{cases}$$

The same bound applies to representations for which g_2 is a proper prime power. Adding the two contributions gives the claim. $\square$

We now analyze $S(a,d)$ through its character decomposition in (2.1). The contribution to $S(a,d)$ from the principal character $\chi = \chi_0$ is

$$M(a,d) := \frac{1}{|B^\times|}\Bigg(\sum_{\substack{\deg g \le d \\ (g,f)=1 \\ g \text{ monic}}} \Lambda(g)\Bigg)^2 = \frac{1}{|B^\times|}\Bigg(\sum_{k\le d}\Bigg(\sum_{\substack{\deg g = k \\ g \text{ monic}}} \Lambda(g) - \sum_{\substack{p\mid f \\ \deg p \mid k \\ p \text{ monic irreducible}}} \deg p\Bigg)\Bigg)^2$$

$$= \frac{1}{|B^\times|}\Bigg(\sum_{k\le d}(q^k + O(\deg f))\Bigg)^2$$

$$= \frac{1}{|B^\times|}\Bigg(\frac{q^{d+1}-q}{q-1} + O(dn)\Bigg)^2$$

$$= q^{2d-n}\Big(1 + O\Big(\tfrac{(dn)^2}{q}\Big)\Big).$$

More precisely, for $q \ge n$, we have

$$M(a,d) \ge \frac{1}{|B^\times|}\Bigg(\sum_{k\le d}(q^k - n)\Bigg)^2 \ge \frac{1}{q^n}\left(q^d - dn\right)^2 = q^{2d-n} - 2dn\, q^{d-n} + \frac{(dn)^2}{q^n}. \tag{2.3}$$

Define

$$R(a,d) := \frac{1}{|B^\times|}\sum_{\substack{\chi \bmod f \\ \chi\ne\chi_0}} \chi(a^{-1})\Bigg(\sum_{\substack{\deg g\le d \\ g \text{ monic}}} \chi(g)\,\Lambda(g)\Bigg)^2.$$

Then

$$S(a,d) = M(a,d) + R(a,d).$$

2.2. **Characters and Dirichlet L-functions.** Let $\chi : B^\times \to \mathbb{C}^\times$ be a multiplicative character. We extend χ to all monic polynomials $h \in k[X]$ by setting

$$\chi(h) := \begin{cases} \chi(\bar h), & \text{if } \bar h \in B^\times, \\ 0, & \text{otherwise,} \end{cases}$$

where $\bar h$ denotes the image of h in B.

Since f is a product of distinct monic irreducible polynomials $f = \prod_{i=1}^r f_i$, every character χ of $B^\times$ decomposes as

$$\chi = \prod_{i=1}^r \chi_i,$$

where χ_i is a character of $B_i^\times$ with $B_i = k[X]/(f_i(X))$.

We say that χ is *primitive* if each character component χ_i is nontrivial. We say that χ is an *odd* character if its restriction to the subgroup $k^\times \subset B^\times$ is nontrivial, and *even* if this restriction is trivial.

We now adopt the terminology from [9].

Definition 2.1. Let k be a finite field and $f \in k[X]$ a squarefree polynomial, and set $B := k[X]/(f(X))$. We say that a character $\chi : B^\times \to \mathbb{C}^\times$ is *totally ramified* if it is both primitive and odd. We denote by $\mathrm{TotRam}(k,f)$ the set of all totally ramified characters of $B^\times$.

The Dirichlet L-function attached to χ is the power series

$$L(\chi,T) := \sum_{\substack{g\in k[X] \\ g \text{ monic}}} \chi(g)\, T^{\deg g}.$$

If $\chi \ne \chi_0$, then $L(\chi,T)$ is a polynomial in T of degree $n-1$. We may factor it as

$$L(\chi,T) = \prod_{i=1}^{n-1}(1-\beta_i T). \tag{2.4}$$

As a consequence of the Riemann Hypothesis for function fields [11, Thm. 5.10], originally proved by Weil [14], each reciprocal root β_i satisfies

$$\text{either } |\beta_i| = 1 \quad \text{or} \quad |\beta_i| = q^{1/2}. \tag{2.5}$$

Alternatively, this can be deduced from Deligne's purity theorem [3] for these Dirichlet L-functions, as shown in [9]. More precisely, when χ is primitive, [9, Lem. 3.5] implies that, if χ is odd, all β_i satisfy $|\beta_i| = q^{1/2}$; and if χ is even, [9, Lem. A.1 and A.2] implies that one of the β_i is 1 and the others satisfy $|\beta_i| = q^{1/2}$. When χ is non-primitive, the additional Euler factors contribute only reciprocal roots of absolute value 1.

For $m \geq 1$, define

$$\Psi(m,\chi) := \sum_{\substack{g \text{ monic} \\ \deg g = m}} \Lambda(g)\,\chi(g).$$

Taking logarithmic derivatives of the Euler product for $L(\chi,T)$ and of the factorization (2.4) gives

$$\Psi(m,\chi) = -\sum_{i=1}^{n-1} \beta_i^m. \tag{2.6}$$

Hence, by (2.5),

$$|\Psi(m,\chi)| \leq (n-1)q^{m/2}, \qquad \chi \neq \chi_0. \tag{2.7}$$

For a totally ramified character χ, each reciprocal root β_i of $L(\chi,T)$ satisfies $|\beta_i| = q^{1/2}$. The unitarized L-function $L(\chi,\ T/q^{1/2})$ is the reversed characteristic polynomial $\det(1 - T\Theta_\chi)$, where

$$\Theta_\chi = \operatorname{diag}\Big(\tfrac{\beta_1}{q^{1/2}}, \dots, \tfrac{\beta_{n-1}}{q^{1/2}}\Big) \in U(n-1).$$

Since conjugacy classes in $U(n-1)$ are determined by their characteristic polynomials, there exists a well-defined conjugacy class

$$\theta_{k,f,\chi} \in U(n-1) \tag{2.8}$$

such that

$$L(\chi,\ T/q^{1/2}) = \det(1 - T\theta_{k,f,\chi}),$$

and by (2.6),

$$\Psi(m,\chi) = -\,q^{m/2}\operatorname{Tr}(\theta_{k,f,\chi}^m).$$

Let $\mathrm{All}(k,f)$ denote the set of all characters on $B^\times$. By [9, Lem. 6.4], we have

$$|\mathrm{All}(k,f)| - |\mathrm{TotRam}(k,f)| \ \leq\ q^n - 1 - (q-2)^n + \sum_{0\leq i\leq n-1} q^i \ \leq\ 2(n{+}1)\,q^{n-1}.$$

Applying (2.7), the contribution from characters $\chi \neq \chi_0$ with $\chi \notin \mathrm{TotRam}(k,f)$ is bounded by

$$\frac{1}{|B^\times|}\,2(n{+}1)\,q^{n-1}\sum_{\substack{i\leq n\\ j\leq n}} n^2 q^{\frac{i+j}{2}} \quad\leq\quad 2(n{+}1)\,n^4\left(\frac{q}{q-1}\right)^n q^{n-1}.$$

Consequently, we may write

$$\begin{aligned} R(a,n) = {} & \frac{1}{|B^\times|}\sum_{\chi\in\mathrm{TotRam}(k,f)} \chi(a^{-1})\sum_{i,j\leq n} q^{\frac{i+j}{2}}\ \operatorname{Tr}(\theta_{k,f,\chi}^i)\operatorname{Tr}(\theta_{k,f,\chi}^j) \\ & + \frac{1}{|B^\times|}\sum_{\substack{\chi\notin\mathrm{TotRam}(k,f)\\ \chi\neq\chi_0}} \chi(a^{-1})\Big(\sum_{m\leq n}\Psi(m,\chi)\Big)^2. \end{aligned}$$

Hence,

(2.9)

$$|R(a,n)| \leq \frac{1}{|B^\times|}\left|\sum_{\chi\in\mathrm{TotRam}(k,f)} \chi(a^{-1})\sum_{i,j\leq n} q^{\frac{i+j}{2}}\operatorname{Tr}(\theta_{k,f,\chi}^i)\operatorname{Tr}(\theta_{k,f,\chi}^j)\right| + 2(n{+}1)n^4\Big(\frac{q}{q-1}\Big)^n q^{n-1}.$$

2.3. **A preliminary exceptional-set bound.** Let $\Delta(d)$ denote the upper bound in Lemma 2.1, so that

$$0 \le S(a,d) - S_p(a,d) \le \Delta(d).$$

Define the set of "bad" elements

$$\mathcal{E}_p := \{\, a \in B^\times : S_p(a,d) = 0 \,\}.$$

If $a \in \mathcal{E}_p$, then $S(a,d) \le \Delta(d)$. When $M(a,d) - \Delta(d) > 0$, we have

$$|R(a,d)| \ge M(a,d) - \Delta(d).$$

Using the orthogonality of characters and the bound (2.7), for $q \ge 4$,

$$\mathbb{E}_a |R(a,d)|^2 := \frac{1}{|B^\times|} \sum_{a \in B^\times} |R(a,d)|^2 \;=\; \frac{1}{|B^\times|^2} \sum_{\substack{\chi \bmod f \\ \chi \ne \chi_0}} \Bigg| \sum_{\substack{\deg g \le d \\ g \text{ monic}}} \chi(g)\,\Lambda(g) \Bigg|^4 \;\le\; \frac{(2n)^4 q^{2d}}{|B^\times|}.$$

For $d \ge n$ and $q \ge n$, the lower bound (2.3) and Lemma 2.1 give

$$M(a,d) - \Delta(d) \ge q^{2d-n}\Big(1 - \frac{10d^2}{q^{d/2}}\Big).$$

Therefore, by Markov's inequality, when $q \ge n$ and $q^{d/2} > 10d^2$ so that $M(a,d) - \Delta(d) > 0$,

$$|\mathcal{E}_p| \le |B^\times| \frac{\mathbb{E}_a |R(a,d)|^2}{(M(a,d) - \Delta(d))^2} \le \frac{(2n)^4 q^{2n-2d}}{\Big(1 - \dfrac{10d^2}{q^{d/2}}\Big)^2}.$$

Hence, for $d > n$, the exceptional set $\mathcal{E}_p$ is empty once q is sufficiently large in terms of d. In the critical case $d = n$, this argument gives only $|\mathcal{E}_p| \le (2n)^4 \Big(1 - \dfrac{10n^2}{q^{n/2}}\Big)^{-2}$.

Since

$$S_p(a,n) \ge M(a,n) - |R(a,n)| - \Delta(n),$$

and

$$\Delta(n) = 8n^2 q^{n/2} = o_n(M(a,n)) \qquad (q \to \infty),$$

in order to prove Theorem 1.1, it suffices to show that uniformly for all $a \in B^\times$,

$$|R(a,n)| = o_n(M(a,n)) \qquad (q \to \infty).$$

For $q \ge 4$, a direct application of (2.7) shows that

$$|R(a,n)| = \Bigg| \frac{1}{|B^\times|} \sum_{\substack{\chi \bmod f \\ \chi \ne \chi_0}} \chi(a^{-1}) \Bigg(\sum_{\substack{\deg g \le n \\ g \text{ monic}}} \chi(g)\,\Lambda(g) \Bigg)^2 \Bigg| \le (2n)^2 q^n,$$

which is insufficient for the desired result. We will restrict to a generic subset of characters for which a $q^{-1/2}$-saving relative to the main term $M(a,n)$ can be obtained.

3. The trace estimate and the proof of the main results

We now state the trace estimate that forms the main input in the proof of Theorem 1.1. Its proof, which relies on Katz's equidistribution theory for finite-field Mellin transforms, is given in Section 4.

Theorem 3.1. *Suppose that $q \ge (n^2 + 4n + 2)^2$. Then for any $a \in B^\times$ and integers $i, j \ge 1$, we have*

$$\Bigg| \sum_{\chi \in \mathrm{TotRam}(k,f)} \chi(a^{-1})\ \mathrm{Tr}(\theta^i_{k,f,\chi})\ \mathrm{Tr}(\theta^j_{k,f,\chi}) \Bigg| \;\le\; (i+j)\sqrt{p(i+j)}\,|\mathrm{TotRam}(k,f)|\ \frac{2(i{+}j{+}1)\, n^{i+j}}{\sqrt{q}},$$

where $p(m)$ denotes the number of partitions of m.

Proof of Theorem 1.1 and Corollary 1.2. We now apply the above estimates to bound $R(a,n)$. Combining (2.9) with Theorem 3.1, when $q \geq (n^2+4n+2)^2$, we obtain

$$|R(a,n)| \leq \sum_{1\leq i,j\leq n} (i+j)\sqrt{p(i+j)}\, q^{\frac{i+j}{2}}\, \frac{2(i+j+1)\, n^{i+j}}{\sqrt{q}} + 2(n+1)n^4\left(\frac{q}{q-1}\right)^n q^{n-1}.$$

Using the classical bound $p(m) \leq e^{\pi\sqrt{2m/3}}$ [1, Thm. 14.5] and that $i+j \leq 2n$,

$$\sum_{1\leq i,j\leq n} (i+j)\sqrt{p(i+j)}\, q^{\frac{i+j}{2}}\, \frac{2(i+j+1)\, n^{i+j}}{\sqrt{q}} \leq 4(2n+1)n^{2n+3}e^{\pi\sqrt{n/3}}q^{n-\frac{1}{2}}.$$

Set

$$C(n) := 4(2n+1)n^{2n+3}e^{\pi\sqrt{n/3}}.$$

Therefore, for all $a \in B^\times$ and $q \geq (n^2+4n+2)^2$,

$$|R(a,n)| \;\leq\; C(n)\, q^{n-\frac{1}{2}} \;+\; 2(n+1)n^4\left(\frac{q}{q-1}\right)^n q^{n-1}.$$

From (2.3), we recall that

$$M(a,n) \geq q^n - 2n^2 + \frac{n^4}{q^n}.$$

Moreover, by Lemma 2.1,

$$S_p(a,n) \geq S(a,n) - \Delta(n) = M(a,n) + R(a,n) - \Delta(n), \qquad \Delta(n) := 8n^2q^{n/2},$$

we obtain

$$S_p(a,n) \geq M(a,n) - |R(a,n)| - \Delta(n).$$

Therefore,

$$S_p(a,n) \geq q^n - 2n^2 + \frac{n^4}{q^n} - C(n)\, q^{n-\frac{1}{2}} - 2(n+1)n^4\left(\frac{q}{q-1}\right)^n q^{n-1} - 8n^2q^{n/2}.$$

When $q^{1/2} \geq C(n) + 4n^2 + 1$, we have $q^n - C(n)q^{n-\frac{1}{2}} \geq (4n^2+1)q^{n-\frac{1}{2}}$ which is larger than the sum of the remaining negative terms in the above lower bound for $S_p(a,n)$. Thus it suffices to take

$$Q(n) := (C(n) + 4n^2 + 1)^2,$$

and whenever $q \geq Q(n)$, we have $S_p(a,n) > 0$ for every $a \in B^\times$.

Alternatively, one may take $Q(n)$ in the simpler form

$$Q(n) = n^{\kappa n}$$

for some absolute constant $\kappa > 0$; for instance, $\kappa = 16$ suffices.

This completes the proof of Theorem 1.1.

To deduce Corollary 1.2, fix $1 \leq r < 2n$ and set $s = 2n - r$. We similarly define

$$S_{r,s}(a) := \frac{1}{|B^\times|}\sum_{\chi \bmod f} \chi(a^{-1})\Bigg(\sum_{\substack{\deg g=r\\ g \text{ monic}}} \chi(g)\Lambda(g)\Bigg)\Bigg(\sum_{\substack{\deg h=s\\ h \text{ monic}}} \chi(h)\Lambda(h)\Bigg).$$

By orthogonality of characters, $S_{r,s}(a)$ is a weighted count of representations of a modulo f as a product of two monic prime powers of degrees r and s. Moreover, by the same argument as in Lemma 2.1, the contribution from representations in which at least one factor is a proper prime power is at most $4n^2q^{n-\frac{1}{2}}$.

The contribution to $S_{r,s}(a)$ from the principal character is

$$M_{r,s}(a) = \frac{1}{|B^\times|}\Bigg(\sum_{\substack{\deg g=r\\ (g,f)=1\\ g \text{ monic}}} \Lambda(g)\Bigg)\Bigg(\sum_{\substack{\deg h=s\\ (h,f)=1\\ h \text{ monic}}} \Lambda(h)\Bigg) \geq \frac{(q^r-n)(q^s-n)}{q^n} \geq q^n - 2nq^{n-1},$$

while the contribution $R_{r,s}(a)$ to $S_{r,s}(a)$ from non-principal characters is bounded by the same quantity as $R(a,n)$. Indeed, in the estimate for $R(a,n)$, only the condition $i+j \leq 2n$ for the indices i, j was used. For $R_{r,s}(a)$, the sum over i, j is replaced by the single pair $(i, j) = (r, s)$, with $r + s = 2n$, so the resulting bound is no larger.

With the same choice of $Q(n)$, $q^n - C(n)q^{n-\frac{1}{2}} - 4n^2 q^{n-\frac{1}{2}} \geq q^{n-\frac{1}{2}}$ which is still larger than the sum of the remaining negative terms. Hence the same choice of $Q(n)$ made above still gives positivity of the corresponding prime count. This proves Corollary 1.2. □

4. Proof of the trace estimate

In this section, we continue to work with a squarefree monic polynomial $f \in k[X]$ of degree $n \geq 2$, and set $B := k[X]/(f(X))$. Let χ be a character of $B^\times$. Choose a prime ℓ that is invertible in k, and fix an embedding $\overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_\ell$. In this way, we regard χ as a $\overline{\mathbb{Q}}_\ell^\times$-valued character of $B^\times$. We begin by recalling the construction of the perverse sheaf $N(\lambda, \chi)$ as in [9].

Let $\mathbb{B}^\times := \mathrm{Res}_{B/k}(\mathbb{G}_{m,B})$, so that $\mathbb{B}^\times(R) = (B \otimes_k R)^\times$, and $i_f : \mathbb{A}^1_k[1/f] \to \mathbb{B}^\times, t \mapsto X - t$. For a character $\chi : B^\times \to \overline{\mathbb{Q}}_\ell^\times$, let $\mathcal{L}_\chi$ be the associated rank-one character sheaf on $\mathbb{B}^\times$. The sheaf $\mathcal{L}_\chi$ is lisse of rank 1 and pure of weight 0.

There is a morphism of algebraic groups $\mathrm{Norm}_{B/k} : \mathbb{B}^\times \to \mathbb{G}_m$ given on R-valued points by the norm maps of the finite étale R-algebras $R[X]/(f)$ restricted to units. These morphisms satisfy

$$\mathrm{Norm}_{B/k} \circ i_f = (-1)^{\deg f} \cdot f, \tag{4.1}$$

see [9, Lem. 3.1].

Denote by $\mathcal{P}$ and $\mathcal{P}_{\mathrm{arith}}$ the Tannakian categories introduced by Katz in [8, Ch. 2]. For $N \in \mathcal{P}_{\mathrm{arith}}$, write $\langle N \rangle_{\mathrm{arith}}$ and $\langle N \rangle_{\mathrm{geom}}$ for the Tannakian subcategories tensor-generated by N in $\mathcal{P}_{\mathrm{arith}}$ and $\mathcal{P}$, respectively.

Definition 4.1. Let N be a perverse sheaf on $\mathbb{G}_{m,k}$, and let ρ be a character of $k^\times$. We say that ρ is *good for* N if, writing $j : \mathbb{G}_{m,\bar{k}} \hookrightarrow \mathbb{P}^1_{\bar{k}}$ for the open immersion, the natural "forget-supports" morphism

$$j_! (N \otimes \mathcal{L}_\rho) \longrightarrow Rj_* (N \otimes \mathcal{L}_\rho)$$

is an isomorphism.

Choose a finite extension E/k over which f splits completely, $f(X) = \prod_{i=1}^n (X - a_i)$. Then the character $\chi_E := \chi \circ \mathrm{Norm}_{E/k}$ of $(B \otimes_k E)^\times \simeq (E^\times)^n$, where $\mathrm{Norm}_{E/k} : (E[X]/(f))^\times \to B^\times$ is the map on units induced by the norm of the finite étale B-algebra $E[X]/(f) \simeq B \otimes_k E$, decomposes uniquely as

$$\chi_E(x_1, \ldots, x_n) = \prod_{i=1}^n \widetilde{\chi}_i(x_i)$$

for characters $\widetilde{\chi}_1, \ldots, \widetilde{\chi}_n$ of $E^\times$. We call $\widetilde{\chi}_1, \ldots, \widetilde{\chi}_n$ the *geometric constituent characters* of χ. The conditions in the theorem below are imposed on these constituent characters.

Theorem 4.1 ([9, Thm. 5.1]). *Let $f \in k[X]$ be a squarefree monic polynomial of degree $n \geq 2$, set $B := k[X]/(f(X))$, and let χ be a character of $B^\times$. Suppose the geometric constituent characters $\widetilde{\chi}_1, \ldots, \widetilde{\chi}_n$ of χ satisfy the following conditions:*

(1) *The characters $\widetilde{\chi}_i$ are pairwise distinct.*
(2) *The product $\prod_i \widetilde{\chi}_i$ is nontrivial (equivalently, χ is nontrivial on $k^\times$).*
(3) *For at least one index i, one has $\widetilde{\chi}_i^n \neq \prod_i \widetilde{\chi}_i$.*

Fix $\lambda \in k^\times$. Then λf defines a k-morphism $\lambda f : \mathbb{A}^1_k[1/f] \to \mathbb{G}_{m,k}$, and we consider the perverse sheaf

$$N(\lambda, \chi) := [\lambda f]_* \left(i_f^* \mathcal{L}_\chi \right) (\tfrac{1}{2})[1]$$

on $\mathbb{G}_{m,k}$.

Then the following properties hold:

(1) *$N(\lambda, \chi)$ is geometrically irreducible, pure of weight 0, and lies in the Tannakian category $\mathcal{P}_{\mathrm{arith}}$. It has generic rank n, Tannakian dimension $n - 1$, and at most $2n$ bad characters.*
(2) *$N(\lambda, \chi)$ is geometrically Lie-irreducible in $\mathcal{P}$.*
(3) *$N(\lambda, \chi)$ has Tannakian groups*

$$G_{\mathrm{geom}, N(\lambda,\chi)} = G_{\mathrm{arith}, N(\lambda,\chi)} = \mathrm{GL}(n - 1).$$

□

We briefly explain the role of the construction of $N(\lambda,\chi)$ and its connection with the L-functions considered in Section 2; further details will appear in the later proofs. For a bounded constructible complex K on $\mathbb{G}_m$, we write $\mathrm{Tr}(\mathrm{Frob}_{k_r,x} \mid K)$ for its trace function at $x \in \mathbb{G}_m(k_r)$, where k_r/k is the extension of degree r. The trace function of $N(\lambda,\chi)$ is

$$\mathrm{Tr}(\mathrm{Frob}_{k,s} \mid N(\lambda,\chi)) = -q^{-1/2} \sum_{\substack{t\in k \\ \lambda f(t)=s}} \chi(X-t), \qquad s \in k^\times.$$

Taking the discrete Mellin transform against a character $\rho : k^\times \to \mathbb{C}^\times$, we obtain

$$\sum_{s\in k^\times} \rho(s)\ \mathrm{Tr}(\mathrm{Frob}_{k,s} \mid N(\lambda,\chi)) = -q^{-1/2} \sum_{t\in \mathbb{A}^1[1/f](k)} \chi(X-t)\rho(\lambda f(t)). \tag{4.2}$$

For a character ρ that is good for $N(\lambda,\chi)$, the Frobenius action on $H^0_c(\mathbb{G}_{m,\bar{k}}, N(\lambda,\chi)\otimes\mathcal{L}_\rho)$ defines a conjugacy class $\theta_{k,\lambda,f,\chi,\rho} \in U(n-1)$, see [8, p. 13]. Since the relevant compactly supported cohomology is concentrated in degree 0, the Grothendieck–Lefschetz trace formula gives

$$\mathrm{Tr}(\theta_{k,\lambda,f,\chi,\rho}) = \mathrm{Tr}\Big(\mathrm{Frob}_k \mid H^0_c(\mathbb{G}_{m,\bar{k}}, N(\lambda,\chi)\otimes\mathcal{L}_\rho)\Big) = \sum_{s\in k^\times} \rho(s)\ \mathrm{Tr}(\mathrm{Frob}_{k,s} \mid N(\lambda,\chi)).$$

In the totally ramified situation, for $\lambda = (-1)^n$, since $(-1)^n f(t) = \mathrm{Norm}_{B/k}(X-t)$, the right-hand side of (4.2) becomes

$$-q^{-1/2} \sum_{t\in \mathbb{A}^1[1/f](k)} (\chi\rho_{\mathrm{Norm}})(X-t) = -q^{-1/2}\Psi(1,\chi\rho_{\mathrm{Norm}}) = \mathrm{Tr}(\theta_{k,f,\chi\rho_{\mathrm{Norm}}}),$$

where $\rho_{\mathrm{Norm}} := \rho\circ\mathrm{Norm}_{B/k}$ and $\theta_{k,f,\chi\rho_{\mathrm{Norm}}}$ is the conjugacy class arising from the corresponding L-function, as defined in (2.8). Hence

$$\mathrm{Tr}(\theta_{k,(-1)^n,f,\chi,\rho}) = \mathrm{Tr}(\theta_{k,f,\chi\rho_{\mathrm{Norm}}}).$$

As will be explained in (4.3), and as shown in [9, p. 3238], one can in fact identify the two conjugacy classes

$$\theta_{k,(-1)^n,f,\chi,\rho} = \theta_{k,f,\chi\rho_{\mathrm{Norm}}}.$$

Therefore, for any representation Λ of $U(n-1)$,

$$\mathrm{Tr}(\Lambda(\theta_{k,f,\chi\rho_{\mathrm{Norm}}})) = \mathrm{Tr}(\Lambda(\theta_{k,(-1)^n,f,\chi,\rho})).$$

The latter traces can then be studied through the Tannakian category generated by $N((-1)^n,\chi)$.

For a character χ of $B^\times$ and $\lambda \in k^\times$, we denote by $\mathrm{Good}(k,\lambda,f,\chi)$ the set of characters $\rho : k^\times \to \mathbb{C}^\times$ that are good for $N(\lambda,\chi)$.

Proposition 4.2. *Let χ be a character of $B^\times$ whose geometric constituent characters $\widetilde{\chi}_i$ satisfy the assumptions of Theorem 4.1. Assume that $q := |k|$ satisfies $\sqrt{q} \geq 2n+1$.*

For each character ρ of $k^\times$ that is good for $N(\lambda,\chi)$ in the sense of Definition 4.1, denote by

$$\theta_{k,\lambda,f,\chi,\rho} \in U(n-1)$$

the conjugacy class whose reversed characteristic polynomial is

$$\det(1 - T\theta_{k,\lambda,f,\chi,\rho}) = \det\Big(1 - T\ \mathrm{Frob}_k \Big| H^0_c(\mathbb{G}_{m,\bar{k}}, N(\lambda,\chi)\otimes\mathcal{L}_\rho)\Big).$$

Let Λ be a nontrivial irreducible representation of $U(n-1)$ contained in $\mathrm{std}^{\otimes a}\otimes(\mathrm{std}^\vee)^{\otimes b}$, and fix $c \in k^\times$. Then the following estimate holds

$$\left| \sum_{\rho\in\mathrm{Good}(k,\lambda,f,\chi)} \rho(c)\ \mathrm{Tr}(\Lambda(\theta_{k,\lambda,f,\chi,\rho})) \right| \leq |\mathrm{Good}(k,\lambda,f,\chi)| \cdot \frac{(2a+2b+1)n^{a+b}}{\sqrt{q}}.$$

Proof. This is a multiplicatively translated variant of [9, Cor. 5.2], which follows from [8, Rem. 7.5] and the proof of [8, Thm. 28.1].

Let

$$t_{c^{-1}} : \mathbb{G}_m \longrightarrow \mathbb{G}_m, \qquad x \longmapsto c^{-1}x$$

be multiplication by c^{-1}. We apply [8, Rem. 7.5] and the proof of [8, Thm. 28.1] to $N := N(\lambda, \chi)$. Let M be the object in $\langle N\rangle_{\mathrm{arith}}$ corresponding to the representation Λ, and set

$$M_c := t_{c^{-1}}^* M.$$

Since M is semisimple, so are $M \otimes \mathcal{L}_\rho$ and $M_c \otimes \mathcal{L}_\rho$, [8, Lem. 2.1] gives

$$H_c^i(\mathbb{G}_{m,\bar{k}}, M \otimes \mathcal{L}_\rho) = H_c^i(\mathbb{G}_{m,\bar{k}}, M_c \otimes \mathcal{L}_\rho) = 0 \qquad (i \neq 0).$$

For any character ρ of $k^\times$, the vanishing above and the Grothendieck–Lefschetz trace formula [7, Exp. III, §4] give

$$\begin{aligned}
\mathrm{Tr}\Big(\mathrm{Frob}_k \,\Big|\, H_c^0(\mathbb{G}_{m,\bar{k}}, M_c \otimes \mathcal{L}_\rho)\Big) &= \sum_{s \in \mathbb{G}_m(k)} \rho(s)\ \mathrm{Tr}(\mathrm{Frob}_{k,s} \mid M_c) \\
&= \sum_{s \in \mathbb{G}_m(k)} \rho(s)\ \mathrm{Tr}(\mathrm{Frob}_{k,c^{-1}s} \mid M) \\
&= \rho(c) \sum_{u \in \mathbb{G}_m(k)} \rho(u)\ \mathrm{Tr}(\mathrm{Frob}_{k,u} \mid M) \\
&= \rho(c)\ \mathrm{Tr}\Big(\mathrm{Frob}_k \,\Big|\, H_c^0(\mathbb{G}_{m,\bar{k}}, M \otimes \mathcal{L}_\rho)\Big).
\end{aligned}$$

In particular, for ρ that is good for $N(\lambda, \chi)$, since

$$\mathrm{Tr}(\Lambda(\theta_{k,\lambda,f,\chi,\rho})) = \mathrm{Tr}\Big(\mathrm{Frob}_k \,\Big|\, H_c^0(\mathbb{G}_{m,\bar{k}},\ M \otimes \mathcal{L}_\rho)\Big),$$

we have

$$\mathrm{Tr}\Big(\mathrm{Frob}_k \,\Big|\, H_c^0(\mathbb{G}_{m,\bar{k}}, M_c \otimes \mathcal{L}_\rho)\Big) = \rho(c)\ \mathrm{Tr}(\Lambda(\theta_{k,\lambda,f,\chi,\rho})).$$

Thus the twisted sum in the proposition is precisely the sum obtained from the translated object M_c. Since $t_{c^{-1}}$ is an automorphism of $\mathbb{G}_m$, the object M_c is again non-punctual and pure of weight 0, with the same generic rank and Tannakian dimension as M. Therefore the argument of [8, Rem. 7.5], together with the proof of [8, Thm. 28.1], applies verbatim with M_c in place of M. Using the same bounds in terms of generic rank and Tannakian dimension as in the proof of [8, Thm. 28.1] yields the claimed estimate. ☐

Remark 4.3. Forey, Fresán, and Kowalski [5] generalize Katz's convolution–equidistribution framework for $\mathbb{G}_m$ to arbitrary connected commutative algebraic groups over finite fields. In particular, one may consider the full torus $\mathbb{B}^\times = \mathrm{Res}_{B/k}\,\mathbb{G}_{m,B}$ whose k-points are $B^\times$, together with the perverse sheaf $M := i_{f*}(\overline{\mathbf{Q}}_\ell[1](1/2))$ arising from the rank-one constant sheaf on $\mathbb{A}_k^1[1/f]$.

From [5, Prop. 4.3], one can obtain twisted character-sum bounds with a $q^{-1/2}$-saving for each fixed object N in the Tannakian category generated by M. The proof shows that the implied constant is controlled by the complexity of N [5, Def. 1.2]. To obtain an estimate analogous to Proposition 4.2 from this framework, one would need to track the complexity of the objects N_Λ corresponding to relevant representations Λ and obtain constants with explicit dependence on the representation parameters. We instead use Katz's one-dimensional Mellin-transform framework on $\mathbb{G}_m$, where these constants are explicit.

Definition 4.2. Let χ be a character of $B^\times$. We say that χ is *totally ramified generic* if

(1) χ is totally ramified in the sense of Definition 2.1, and
(2) the geometric constituent characters $\widetilde{\chi}_i$ of χ satisfy the three conditions of Theorem 4.1.

We denote by $\mathrm{TotRamGen}(k, f)$ the set of totally ramified generic characters of $B^\times$.

We first record the compatibility between the conjugacy classes appearing in Proposition 4.2 and those defined in (2.8), following the discussion in [9, pp. 3226–3228, 3238]. By [9, Lem. 2.1],

$L(\chi,T)=\exp\Big(\sum_{r\geq 1}\frac{S_rT^r}{r}\Big)$, where

$$\begin{aligned}S_r&=\sum_{t\in\mathbb{A}^1[1/f](k_r)}\chi(\mathrm{Norm}_{k_r/k}(X-t))\\&=\sum_{t\in\mathbb{A}^1[1/f](k_r)}\mathrm{Tr}(\mathrm{Frob}_{k_r,t}\mid i_f^*\mathcal{L}_\chi)\\&=\sum_{i=1,2}(-1)^i\,\mathrm{Tr}(\mathrm{Frob}_k^r\mid H_c^i(\mathbb{A}^1[1/f]_{\bar{k}},\,i_f^*\mathcal{L}_\chi)).\end{aligned}$$

Therefore,

$$L(\chi,T)=\frac{\det\Big(1-T\ \mathrm{Frob}_k\mid H_c^1(\mathbb{A}^1[1/f]_{\bar{k}},\,i_f^*\mathcal{L}_\chi)\Big)}{\det\Big(1-T\ \mathrm{Frob}_k\mid H_c^2(\mathbb{A}^1[1/f]_{\bar{k}},\,i_f^*\mathcal{L}_\chi)\Big)}.$$

For $\chi\in\mathrm{TotRamGen}(k,f)$, by [9, Lem. 3.2] we have $H_c^2(\mathbb{A}^1[1/f]_{\bar{k}},\,i_f^*\mathcal{L}_\chi)=0$, and thus

$$L(\chi,T)=\det\Big(1-T\ \mathrm{Frob}_k\mid H_c^1(\mathbb{A}^1[1/f]_{\bar{k}},\,i_f^*\mathcal{L}_\chi)\Big).$$

Hence the Leray spectral sequence gives

$$\begin{aligned}H_c^0(\mathbb{G}_{m,\bar{k}},\,N(\lambda,\chi)\otimes\mathcal{L}_\rho)&\cong H_c^0(\mathbb{A}^1[1/f]_{\bar{k}},\,i_f^*\mathcal{L}_\chi\otimes[\lambda f]^*\mathcal{L}_\rho(1/2)[1])\\&\cong H_c^1(\mathbb{A}^1[1/f]_{\bar{k}},\,i_f^*\mathcal{L}_\chi\otimes[\lambda f]^*\mathcal{L}_\rho(1/2)).\end{aligned}$$

When $\lambda=(-1)^n$, by (4.1), we have $\mathrm{Norm}_{B/k}\circ i_f=(-1)^nf$. Denote by ρ_{Norm} the character of $B^\times$ defined by $\rho_{\mathrm{Norm}}:=\rho\circ\mathrm{Norm}_{B/k}$. Then $[(-1)^nf]^*\mathcal{L}_\rho=i_f^*(\mathrm{Norm}_{B/k}^*\mathcal{L}_\rho)=i_f^*\mathcal{L}_{\rho_{\mathrm{Norm}}}$. Therefore, the conjugacy class $\theta_{k,(-1)^n,f,\chi,\rho}$ appearing in Proposition 4.2 coincides with the conjugacy class $\theta_{k,f,\chi\rho_{\mathrm{Norm}}}$ defined in (2.8)

$$\theta_{k,(-1)^n,f,\chi,\rho}=\theta_{k,f,\chi\rho_{\mathrm{Norm}}}. \tag{4.3}$$

Proposition 4.4. *Suppose $\sqrt{q}\geq 2n+1$. Let Λ be a nontrivial irreducible representation of $U(n-1)$ that occurs in $\mathrm{std}^{\otimes a}\otimes(\mathrm{std}^\vee)^{\otimes b}$, and let $c\in B^\times$. Then the following estimate holds*

$$\Bigg|\sum_{\chi\in\mathrm{TotRamGen}(k,f)}\chi(c)\ \mathrm{Tr}\big(\Lambda(\theta_{k,f,\chi})\big)\Bigg|\ \leq\ |\mathrm{TotRamGen}(k,f)|\cdot\frac{(2a+2b+1)\,n^{a+b}}{\sqrt{q}}.$$

Proof. Following [9, Cor. 6.2], partition $\mathrm{TotRamGen}(k,f)$ into equivalence classes, where two characters χ,χ' are equivalent if $\chi'=\chi\,\rho_{\mathrm{Norm}}$ for some character $\rho\in\mathrm{Good}(k,(-1)^n,f,\chi)$. For a fixed representative χ, its equivalence class is

$$\{\chi\rho_{\mathrm{Norm}}:\rho\in\mathrm{Good}(k,(-1)^n,f,\chi)\}.$$

Therefore, applying Proposition 4.2 to each equivalence class, we obtain

$$\begin{aligned}&\Bigg|\sum_{\chi\in\mathrm{TotRamGen}(k,f)}\chi(c)\ \mathrm{Tr}\big(\Lambda(\theta_{k,f,\chi})\big)\Bigg|\\&=\Bigg|\sum_{[\chi]\in\mathrm{TotRamGen}(k,f)/\sim}\chi(c)\sum_{\rho\in\mathrm{Good}(k,(-1)^n,f,\chi)}\rho(\mathrm{Norm}_{B/k}(c))\ \mathrm{Tr}\big(\Lambda(\theta_{k,f,\chi\rho_{\mathrm{Norm}}})\big)\Bigg|\\&=\Bigg|\sum_{[\chi]\in\mathrm{TotRamGen}(k,f)/\sim}\chi(c)\sum_{\rho\in\mathrm{Good}(k,(-1)^n,f,\chi)}\rho(\mathrm{Norm}_{B/k}(c))\ \mathrm{Tr}\big(\Lambda(\theta_{k,(-1)^n,f,\chi,\rho})\big)\Bigg|\\&\leq|\mathrm{TotRamGen}(k,f)|\cdot\frac{(2a+2b+1)\,n^{a+b}}{\sqrt{q}}.\end{aligned}$$

∎

Proof of Theorem 3.1. Suppose that $q \geq (n^2+4n+2)^2$, we now show that for any $a \in B^\times$ and integers $i, j \geq 1$,

$$\left| \sum_{\chi \in \mathrm{TotRam}(k,f)} \chi(a^{-1}) \, \mathrm{Tr}(\theta^i_{k,f,\chi}) \, \mathrm{Tr}(\theta^j_{k,f,\chi}) \right| \leq (i+j)\sqrt{p(i+j)}\, |\mathrm{TotRam}(k,f)| \, \frac{2(i+j+1)\, n^{i+j}}{\sqrt{q}},$$

where $p(m)$ denotes the number of partitions of m. We may assume that $i \leq j$.

We split the sum into the contributions from $\chi \in \mathrm{TotRamGen}(k,f)$ and from the remaining characters in $\mathrm{TotRam}(k,f) \setminus \mathrm{TotRamGen}(k,f)$.

We first consider the contribution from the former set. Recall the power-sum symmetric functions

$$P_j(x_1, x_2, \ldots, x_{n-1}) = \sum_{i=1}^{n-1} x_i^j,$$

and for a partition $\lambda = 1^{a_1} 2^{a_2} \cdots k^{a_k}$, define

$$P_\lambda = \prod_{j=1}^{k} P_j^{a_j}.$$

Here λ is a partition of $K = a_1 + 2a_2 + \cdots + ka_k$. As λ ranges over all partitions of K, the functions P_λ form a basis for the space of homogeneous symmetric polynomials of degree K with rational coefficients; see [6, pp. 459–460].

A second basis for this space is given by the Schur functions s_μ, as defined in [6, (A.4)]; see also [6, (A.17), (A.23)] for the basis property. They are related to the power sums by the Frobenius character formula

$$P_\lambda = \sum_{\mu \vdash K} \chi_\mu(\lambda)\, s_\mu,$$

where $\chi_\mu(\lambda)$ denotes the value of the irreducible character of S_K associated to μ, evaluated on the conjugacy class of cycle type λ; see [6, p. 49 and pp. 54-57].

In our setting, we consider $P_{i,j}$ since $\mathrm{Tr}(\theta^i_{k,f,\chi})\,\mathrm{Tr}(\theta^j_{k,f,\chi})$ is the evaluation of $P_{i,j}$ on the eigenvalues of $\theta_{k,f,\chi}$. For $\lambda = (i,j)$ we have

$$P_{i,j} = \sum_{\mu \vdash i+j} \chi_\mu(i,j)\, s_\mu.$$

Let $\mathbb{S}_\mu(\mathbb{C}^{n-1})$ denote the Weyl module associated with μ; see [6, pp. 76-77]. If μ has length at most $n-1$, then it is an irreducible representation of $\mathrm{GL}_{n-1}(\mathbb{C})$ (and hence of $U_{n-1}(\mathbb{C})$), with character given by the Schur function s_μ; otherwise it is zero; see [6, Thm. 6.3].

By the Cauchy–Schwarz inequality and the column orthogonality of characters [6, Ex. 2.21(i)], we have

$$\begin{aligned} \sum_{\mu \vdash i+j} |\chi_\mu(i,j)| &\leq \sqrt{p(i+j)} \Big(\sum_{\mu \vdash i+j} |\chi_\mu(i,j)|^2 \Big)^{1/2} \\ &= \begin{cases} \sqrt{p(i+j)}\, \sqrt{ij}, & i \neq j, \\ \sqrt{p(i+j)}\, \sqrt{2i^2}, & i = j. \end{cases} \end{aligned}$$

In particular,

$$\sum_{\mu \vdash i+j} |\chi_\mu(i,j)| \leq (i+j)\sqrt{p(i+j)}. \tag{4.4}$$

Therefore, applying Proposition 4.4 to each $\mathbb{S}_\mu(\mathbb{C}^{n-1})$, summing over the partitions $\mu \vdash i+j$, and using (4.4) above, we obtain, for any $a \in B^\times$,

$$\left| \sum_{\chi \in \mathrm{TotRamGen}(k,f)} \chi(a^{-1}) \, \mathrm{Tr}(\theta^i_{k,f,\chi}) \, \mathrm{Tr}(\theta^j_{k,f,\chi}) \right| \leq (i+j)\sqrt{p(i+j)}\, |\mathrm{TotRamGen}(k,f)| \cdot \frac{(2i+2j+1)\, n^{i+j}}{\sqrt{q}}.$$

For the second part, the sum has at most $|\mathrm{TotRam}(k,f)| - |\mathrm{TotRamGen}(k,f)|$ terms. Since $\theta_{k,f,\chi}$ is a unitary conjugacy class, $\left|\mathrm{Tr}(\theta^i_{k,f,\chi})\,\mathrm{Tr}(\theta^j_{k,f,\chi})\right| \leq (n-1)^2$. We will combine this trivial bound with an upper bound for $|\mathrm{TotRam}(k,f)| - |\mathrm{TotRamGen}(k,f)|$ to get an upper bound for the second part.

As shown in [9, Lem. 6.5–6.6], when $q \geq n+1$ we have

$$|\mathrm{TotRamGen}(k,f)| \geq P_{\mathrm{TRG},n}(q),$$
$$|\mathrm{TotRam}(k,f)| - |\mathrm{TotRamGen}(k,f)| \leq q^n - 1 - P_{\mathrm{TRG},n}(q),$$

where $P_{\mathrm{TRG},n}(q)$ denotes the polynomial

$$P_{\mathrm{TRG},n}(q) = (q-1-n)^n + (q-2)^n - q^n + 1 - n \sum_{0 \leq i \leq n-1} q^i.$$

When $q \geq n+1$, we have $q^n - 1 - P_{\mathrm{TRG},n}(q) \leq (n^2+4n+1)q^{n-1}$. Thus, if $q \geq (n^2+4n+2)^2$, then $q^n - 1 - P_{\mathrm{TRG},n}(q) \leq \dfrac{P_{\mathrm{TRG},n}(q)}{\sqrt{q}}$, it follows that

$$|\mathrm{TotRam}(k,f)| - |\mathrm{TotRamGen}(k,f)| \;\leq\; \frac{|\mathrm{TotRamGen}(k,f)|}{\sqrt{q}} \;\leq\; \frac{|\mathrm{TotRam}(k,f)|}{\sqrt{q}}.$$

Note that the condition $q \geq n+1$ is implied by $q \geq (n^2+4n+2)^2$. Combining the two estimates yields the desired bound. □

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